Sustainability Aware Asset Management

Lecture 1: General Problem of Asset Management

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Objectives of the lecture

- **What Is Sustainable Finance?**
- Portfolio Optimization
- Capital Asset Pricing Model
- Market Efficiency

Why Sustainable Finance Matters (Even in Tough Times)

- **Mandates and constraints are structural:** regulation, client preferences, labels, and reporting now shape investable universes.
- **Risk is shifting:** transition policy, technology, litigation, and physical risks affect cash flows and discount rates.
- **Data is messy (and valuable):** ESG/climate signals are noisy and disagree across providers — which creates room for skill.
- **Capital allocation is changing:** the transition re-ranks sectors and firms; understanding it is part of modern asset management.

Sustainable finance

It is not a belief system; it is **portfolio engineering under new constraints**.

Where This Shows Up in Real Jobs

- **Asset management / indexing:** climate-aware indices, net-zero benchmarks, tracking-error optimization.
- **Risk management:** scenario analysis, factor exposures, stress tests, model risk from ESG data.
- **Banking / credit:** transition risk in spreads, covenant design, sustainability-linked instruments.
- **Consulting / data:** ESG analytics, disclosure mapping, controversy monitoring, metric construction.

Core skill

Translate qualitative objectives into **quantitative, auditable** portfolio rules.

What You Will Be Able To Do After This Course

1. **Frame objectives:** values vs impact, climate vs ESG, exclusion vs engagement.
2. **Measure exposures:** footprints, intensities, alignment metrics, and their limitations.
3. **Build portfolios:** benchmark-relative constraints, tracking error control, robust optimization.
4. **Evaluate outcomes:** performance, risk, unintended tilts, and "impact" claims.

Bottom line

You will learn to **design and defend** sustainability-aware investment processes.

Roadmap: How the Course Answers These Questions

- **Quantitative foundations (Lectures 1-5):** benchmarking, estimation risk, robust portfolio construction, implementation.
- **Climate investing (Lectures 6-9):** measure risk, decarbonize portfolios, hedge carbon risk, net-zero trajectories.
- **ESG investing (Lectures 10-12):** ESG data and strategies, engagement, environment/impact measurement.
- **Institutional layer (Lecture 13):** regulation and sustainable products.

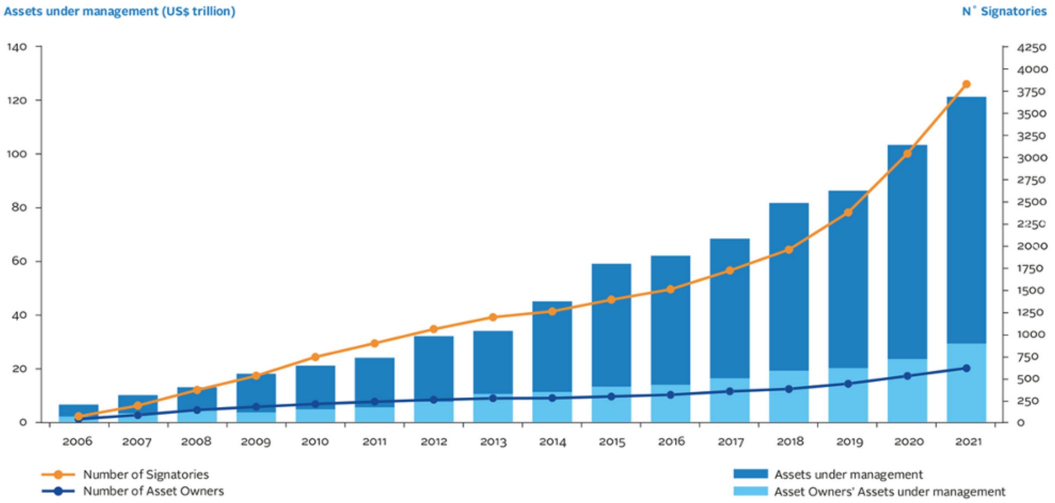
What Is Sustainable Finance?

Sustainable finance: investment decision-making that **integrates** environmental, social, and governance (ESG) considerations as **constraints, risks, and objectives** — with the aim of improving decision quality over relevant horizons.

Operational view: **portfolio choice under additional constraints and noisy measurements**.

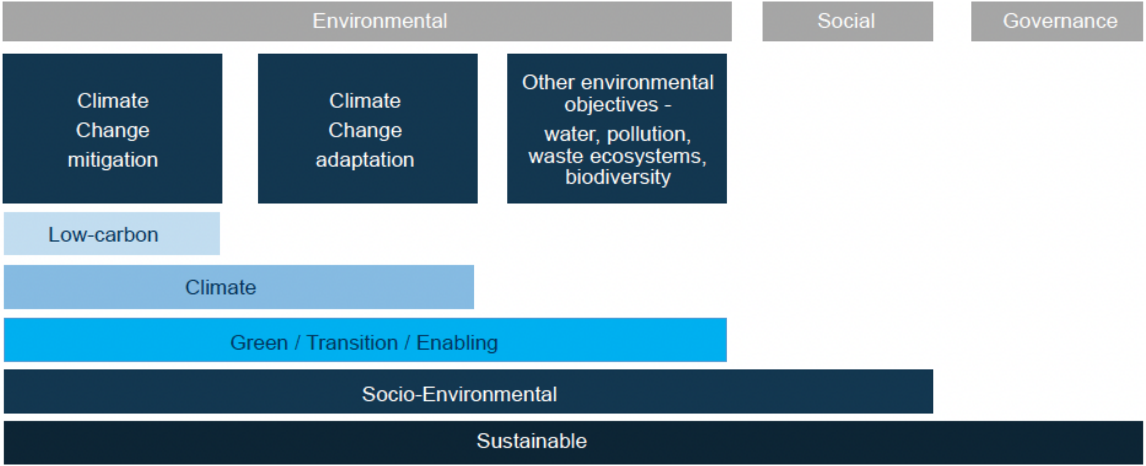
- **Environmental:** climate (physical/transition), biodiversity, pollution, circular economy.
- **Social:** labor, inequality, inclusiveness, human capital, human rights.
- **Governance:** incentives, controls, remuneration, accountability; enables E & S integration.

How Big Is Sustainable Finance?



Source: `https://www.unpri.org/about-us/about-the-pri` (not available anymore)

Sustainable Finance Taxonomy



Source: Sustainable finance hub, "SDG Impact Management and Finance Tracking"

Examples of Sustainable Finance

- **Excluding firms** that do not satisfy certain sustainable criteria
  - e.g., firms violating human rights, causing severe environmental damage, or involved in weapons of mass destruction
- **Engagement with firms** to change corporate policies
  - e.g., meetings with management; voting on shareholder proposals at AGMs
- **Green bonds:** bonds issued to finance environmentally friendly projects (renewable energy, conservation).
- **Impact investing:** investing in companies/organizations/projects with positive social or environmental impact.
- **Sustainable banking:** environmentally and socially responsible policies in operations and lending.

Challenges and Opportunities

- **Challenges:** Transitioning to a more sustainable financial system will require changes in regulation, market infrastructure, and investor behavior
- **Greenwashing:** Process of conveying a false impression or misleading information about how a company's products are environmentally sound. Greenwashing involves making an unsubstantiated claim to deceive consumers into believing that a company's products are environmentally friendly or have a greater positive environmental impact than they actually do (Investopedia)
- **Opportunities:** Sustainable finance also offers opportunities for innovation, risk management, and long-term value creation

Several debates about ESG and climate investing

- ESG investing versus climate investing
- Values-aligned preferences versus impact-seeking preferences
- Exclusion versus engagement
- Impact on climate versus impact on firms
- Cost of exclusion versus price impact

ESG investing versus Climate investing

ESG investing

Consider all non-financial factors:

- Environment
- Social
- Governance

$\Rightarrow$ **Everything is mixed in a single measure**

Climate investing

Focus on:

- Climate change
- Planetary boundaries
- Biodiversity

$\Rightarrow$ **Question: can we fix all problems with climate only?**

Values versus Impact

Values-alignment preferences

Aversion to owning shares of companies not aligned with moral values

$\Rightarrow$ **Adoption of social norms (no alcohol, tobacco, weapons, gambling – sin stocks)**

$\Rightarrow$ **Exclusion of carbon-intensive firms**

Impact-seeking preferences

Investors value social consequences of their investment decisions (additionality)

$\Rightarrow$ **Engagement with carbon-intensive firms**

$\Rightarrow$ **Investment in green start-ups**

Exclusion versus Engagement

Exclusion approach

- Investors renounce impact on excluded firms' decisions
- Low climate risk but possibly large tracking error

$\Rightarrow$ **Large impact on investors' portfolio (green portfolio)**

$\Rightarrow$ **Limited impact on society (global emissions)**

Engagement with firms

- Investors remain invested and try to influence firms
- High climate risk but close to standard benchmark

$\Rightarrow$ **Limited impact on investors' portfolio (close to benchmark)**

$\Rightarrow$ **Potentially large impact on society (global emissions)**

Impact on climate versus Impact on firms

Impact of firms on climate

- Responsibility of firms and investors relative to climate change
- Asset management question

$\Rightarrow$ **Exclusion?**

$\Rightarrow$ **Use of physical measure of climate risk**

Impact of climate on firms

- Firms and investors are exposed to climate risk
- Risk management question

$\Rightarrow$ **Transition risk**

$\Rightarrow$ **Use of market measure of climate risk**

Cost of exclusion versus price impact

Cost of exclusion strategy

Green investors want to do well while doing good

$\Rightarrow$ **Ideally: benefit from exclusion if transition risk materializes**

Price impact of exclusion strategy

Theory: higher premium (cost of capital) to brown firms

$\Rightarrow$ **Green investors should under-perform**

Is there a price impact?

- **NO:** green investors do not under-perform but have limited impact
- **YES:** green investors have an impact (higher cost of capital) but under-perform

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Major Steps in Asset Management

- **Optimal portfolio allocation** (Markowitz, 1952): optimization reduces to a mean-variance criterion.
- **Capital Asset Pricing Model (CAPM)** (Sharpe, 1964; Lintner, 1965): at equilibrium, one factor (beta) explains cross-sectional differences in expected returns.
- **Market efficiency** (Fama, 1965): if markets are efficient, abnormal returns should be unpredictable.

We now discuss these 3 steps.

Major Steps in Asset Management

What is the optimal decision of individual investors?

What makes a difference between stock returns today?

Can we predict stock returns for tomorrow?

Optimal portfolio allocation

(Optimal weights given expected excess returns)

$\xrightarrow{\text{Equilibrium}}$

Capital Asset Pricing Model / Factor Models

(Cross-section heterogeneity)

μ_i^e for all firms

Efficient market hypothesis

(Time-series dynamics)

Unexpected returns $\varepsilon_{i,t+1}$ are unpredictable

Complete description of excess returns properties

$$R_{i,t+1} - R_{f,t} = \mu_i^e + \varepsilon_{i,t+1}$$

Utility Function

An individual with current wealth W_t decides:

- how much wealth to consume now,
- how to invest remaining wealth in financial assets for future consumption.

Asset allocation focuses on the second decision.

Investment objectives are determined by a utility function $U(\cdot)$.

Three premises:

1. Investors prefer more to less: utility increases with wealth.
2. Investors prefer more to less at a decreasing rate: utility is concave in wealth.
3. Risky wealth distributions are valued by the certainty equivalent of expected utility.

Utility Function

Constant absolute risk aversion (CARA)

- Coefficient of absolute risk aversion: $a(W) = -\dfrac{U''(W)}{U'(W)}$
- **Negative exponential utility (CARA):** $U(W) = -\exp(-\lambda W)$
- The coefficient of absolute risk aversion $a(W) = \lambda > 0$ is constant, i.e., independent from initial wealth.

Constant relative risk aversion (CRRA)

- Coefficient of relative risk aversion: $\rho(W) = -W\dfrac{U''(W)}{U'(W)} = Wa(W)$
- **Power utility (CRRA):** $U(W) = \dfrac{W^{1-\gamma}}{1-\gamma}$
- The relative risk aversion $\rho(W) = \gamma > 0$ is constant, i.e., independent from initial wealth.
- Remark: $\gamma = 1$ corresponds to logarithmic utility $U(W) = \log W$.

Single-Period Optimization Problem

Investor maximizes expected utility of next-period wealth: $\mathbb{E}_t\left[U(W_{t+1})\right]$

- N assets with vector of simple returns $\mathbf{R}_{t+1} = \left(R_{1,t+1}, \ldots \cdots, R_{N,t+1}\right)'$.
- No costs for short selling.
- Normalize beginning-of-period wealth: $W_t = 1$.
- $\boldsymbol{\alpha}_t$: fractions of wealth allocated to each risky asset, with budget constraint $\mathbf{e}'\boldsymbol{\alpha}_t = 1$.

Then next-period wealth and portfolio return are:

$$W_{t+1} = W_t\left(1 + R_{p,t+1}\right) = \left(1 + \boldsymbol{\alpha}_t'\mathbf{R}_{t+1}\right), \qquad R_{p,t+1}(\boldsymbol{\alpha}_t) = \boldsymbol{\alpha}_t'\mathbf{R}_{t+1}.$$

If a risk-free asset exists at rate R_f and investors can borrow/lend:

$$R_{p,t+1}(\boldsymbol{\alpha}_t) = R_{f,t} + \boldsymbol{\alpha}_t'(\mathbf{R}_{t+1} - R_{f,t}\mathbf{e}),$$

Single-Period Optimization Problem

Optimal portfolio weights maximize the scaled expected utility

$$\boldsymbol{\alpha}_t^* = \arg\max_{\{\boldsymbol{\alpha}_t\}} \mathbb{E}_t\left[U(W_{t+1})\right].$$

or solve first-order conditions:

$$\frac{\partial}{\partial \boldsymbol{\alpha}_t} \mathbb{E}_t\left[U(W_{t+1})\right] = \mathbb{E}_t\left[U^{(1)}(W_{t+1}) \times (R_{t+1} - R_{f,t}\mathbf{e})\right] = 0$$

- System of N nonlinear equations in N unknowns
- Requires expectations over the joint distribution of returns (N-dimensional integral)
- No general analytical solution
- Numerical solutions feasible for small N, challenging for large N
- Practical bottleneck: forecasting expected returns and the covariance matrix

Mean-Variance Optimization

Harry Markowitz
Nobel Prize (1990)

James Tobin
Nobel Prize (1981)

Notations

Assume N risky securities.

- Returns: $\mathbf{R}_{t+1} = \left(R_{1,t+1}, \ldots R_{N,t+1}\right)'$
- Expected returns (at time t): $\boldsymbol{\mu}_t = \mathbb{E}_t[\mathbf{R}_{t+1}] = (\mu_{1,t}, \cdots, \mu_{N,t})'$
- Covariance matrix: $\boldsymbol{\Sigma}_t = \mathrm{Var}_t(\mathbf{R}_{t+1}) = \begin{pmatrix} \sigma^2_{1,t} & \cdots & \sigma_{1N,t} \\ \vdots & \ddots & \vdots \\ \sigma_{N1,t} & \cdots & \sigma^2_{N,t} \end{pmatrix}$
- Portfolio weights: $\boldsymbol{\alpha}_t = \left(\alpha_{1,t}, \cdots, \alpha_{N,t}\right)'$ with $\mathbf{e}'\boldsymbol{\alpha}_t = 1$
- Ex-post portfolio return: $R_{p,t+1} = \boldsymbol{\alpha}'_t\, \mathbf{R}_{t+1}$
- Portfolio expected return: $\mu_{p,t} = \boldsymbol{\alpha}'_t\, \boldsymbol{\mu}_t$
- Ex-ante portfolio variance: $\sigma^2_{p,t} = \boldsymbol{\alpha}'_t \boldsymbol{\Sigma}_t \boldsymbol{\alpha}_t$

Mean-Variance Case – The exact case

The nice case: exponential utility and normal returns. The exponential utility is

$$U(W_{t+1}) = -\exp(-\lambda W_{t+1}), \qquad \lambda \geq 0 \text{ (absolute risk aversion)}.$$

Let $\boldsymbol{\mu}_t$ be the vector of expected returns and $\boldsymbol{\Sigma}_t$ the covariance matrix. If $X \sim \mathcal{N}(a, b^2)$, then $\mathbb{E}[\exp(X)] = \exp\left(a + \frac{b^2}{2}\right)$.

Then, the optimization problem can be rewritten as:

$$\mathbb{E}_t\left[-\exp(-\lambda W_{t+1})\right] = \mathbb{E}_t\left[-\exp\left(-\lambda(1 + \boldsymbol{\alpha}_t'\mathbf{R}_{t+1})\right)\right] = -\exp\left(-\lambda(1 + \mu_{p,t}) + \frac{1}{2}\lambda^2\sigma_{p,t}^2\right)$$

where $\mu_{p,t} = \boldsymbol{\alpha}_t'\boldsymbol{\mu}_t$ and $\sigma_{p,t}^2 = \boldsymbol{\alpha}_t'\boldsymbol{\Sigma}_t\boldsymbol{\alpha}_t$.

Maximizing $\mathbb{E}_t[U(W_{t+1})]$ is equivalent to maximizing $\mu_{p,t} - \frac{\lambda}{2}\sigma_{p,t}^2$

Mean-Variance Case – The approximate case

Another approach is to express the expected utility function as a Taylor series expansion around $\bar{W} = W_t(1 + R_{f,t})$. The utility function is approximated by an expansion up to the second order:

$$U(W_{t+1}) = U(\bar{W}) + U^{(1)}(\bar{W})(W_{t+1} - \bar{W}) + \frac{1}{2}U^{(2)}(\bar{W})(W_{t+1} - \bar{W})^2 + \varepsilon.$$

so that the expected utility is simply approximated by

$$\begin{aligned}\mathbb{E}_t[U(W_{t+1})] &\approx U(\bar{W}) + U^{(1)}(\bar{W})\,\mathbb{E}_t[W_{t+1} - \bar{W}] + \frac{1}{2}U^{(2)}(\bar{W})\,\mathbb{E}_t\left[(W_{t+1} - \bar{W})^2\right] \\ &\approx U(\bar{W}) + U^{(1)}(\bar{W})\,W_t\,\mathbb{E}_t[R_{p,t+1} - R_{f,t}] + \frac{1}{2}U^{(2)}(\bar{W})\,W_t^2\,\sigma_{p,t}^2.\end{aligned}$$

Maximizing $\mathbb{E}_t[U(W_{t+1})]$ is equivalent to maximizing $(\mu_{p,t} - R_{f,t}) - \frac{\lambda}{2}\sigma_{p,t}^2$, where $\lambda = -\,W_t\,\frac{U^{(2)}(\bar{W})}{U^{(1)}(\bar{W})}$ is the relative risk aversion parameter.

Remark: The mean-variance criterion completely describes $\mathbb{E}[U]$ only for 2-parameter distributions such as the Gaussian or the Uniform distributions.

Mean-Variance Case (Markowitz, 1952) – Solution without a risk-free asset

Are we indifferent between all these portfolios? It depends on our **risk aversion** and therefore on how we trade-off risk and return.

We denote by λ the risk aversion parameter.

When there is no risk-free asset, we must take care of the constraint $\boldsymbol{\alpha}_t'\mathbf{e} = 1$.

The solution is the portfolio weight $\boldsymbol{\alpha}_t^*$ that maximizes:

$$\begin{aligned} \max_{\{\alpha_t\}} \quad & \mu_{p,t} - \frac{\lambda}{2}\sigma_{p,t}^2 = \boldsymbol{\alpha}_t'\boldsymbol{\mu}_t - \frac{\lambda}{2}\boldsymbol{\alpha}_t'\boldsymbol{\Sigma}_t\boldsymbol{\alpha}_t \\ \text{s.t.} \quad & \boldsymbol{\alpha}_t'\mathbf{e} = 1 \end{aligned}$$

γ: Lagrange multiplier.

From now on, we use $\boldsymbol{\alpha} = \boldsymbol{\alpha}_t$, $\boldsymbol{\mu} = \boldsymbol{\mu}_t$, $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_t$

Mean-Variance Case (Markowitz, 1952) – Solution without a risk-free asset

The Lagrangian is: $\mathcal{L} = \boldsymbol{\alpha}'\boldsymbol{\mu} - \frac{\lambda}{2}\boldsymbol{\alpha}'\boldsymbol{\Sigma}\boldsymbol{\alpha} - \gamma(\boldsymbol{\alpha}'\mathbf{e} - 1)$

The derivatives are:

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\alpha}} = \boldsymbol{\mu} - \lambda\boldsymbol{\Sigma}\boldsymbol{\alpha} - \gamma\mathbf{e} = 0, \qquad \text{and} \qquad \frac{\partial \mathcal{L}}{\partial \gamma} = \boldsymbol{\alpha}'\mathbf{e} - 1 = 0.$$

Step 1: $\boldsymbol{\mu} - \lambda\boldsymbol{\Sigma}\boldsymbol{\alpha} - \gamma\mathbf{e} = 0 \quad \Rightarrow \quad \boldsymbol{\alpha} = \frac{1}{\lambda}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \gamma\mathbf{e})$

Step 2: $\mathbf{e}'\boldsymbol{\alpha} = \frac{1}{\lambda}\mathbf{e}'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \gamma\mathbf{e}) = 1 \quad \Rightarrow \quad \mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - \gamma\,\mathbf{e}'\boldsymbol{\Sigma}^{-1}\mathbf{e} = \lambda$

$$\Rightarrow \gamma = \frac{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - \lambda}{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\mathbf{e}}$$

Step 3: $\boldsymbol{\alpha} = \frac{1}{\lambda}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - \frac{1}{\lambda}\boldsymbol{\Sigma}^{-1}\mathbf{e}\,\frac{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - \lambda}{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\mathbf{e}}$

$$\Rightarrow \boldsymbol{\alpha} = \frac{1}{\lambda}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \frac{\boldsymbol{\Sigma}^{-1}\mathbf{e}}{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\mathbf{e}} - \frac{1}{\lambda}\boldsymbol{\Sigma}^{-1}\mathbf{e}\,\frac{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\mathbf{e}}$$

Mean-Variance Case (Markowitz, 1952) – Solution without a risk-free asset

Proposition: When there is no risk-free asset and no restriction on portfolio weights, the weights of the optimal portfolio are:

$$\boldsymbol{\alpha}^* = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{e}}{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\mathbf{e}} + \frac{1}{\lambda}\boldsymbol{\Sigma}^{-1}\left(\boldsymbol{\mu} - \frac{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\mathbf{e}}\,\mathbf{e}\right) = \boldsymbol{\alpha}^*_{\text{gmv}} + \frac{1}{\lambda}\boldsymbol{\Sigma}^{-1}\left(\boldsymbol{\mu} - \frac{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\mathbf{e}}\,\mathbf{e}\right),$$

where $\boldsymbol{\alpha}_{\text{gmv}}$ is independent from expected returns.

$$\boldsymbol{\alpha}^* = \left(1 - \frac{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\lambda}\right)\boldsymbol{\alpha}^*_{\text{gmv}} + \left(\frac{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\lambda}\right)\boldsymbol{\alpha}^*_{\text{spec}}.$$

We obtain the well-known **Mutual Fund Separation Theorem**. Investors invest in:

- the **global minimum-variance portfolio**: $\boldsymbol{\alpha}^*_{\text{gmv}} = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{e}}{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\mathbf{e}}$ with weight $\left(1 - \frac{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\lambda}\right)$
- the **speculative portfolio**: $\boldsymbol{\alpha}^*_{\text{spec}} = \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}$ with weight $\left(\frac{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\lambda}\right)$

An investor minimizing the portfolio variance subject to a return constraint or maximizing the portfolio return subject to a volatility constraint will find the same solution (with no particular value for the risk aversion parameter):

$$\begin{cases} \min\limits_{\{\alpha\}} & \boldsymbol{\Sigma}_p^2 = \boldsymbol{\alpha}'\boldsymbol{\Sigma}\boldsymbol{\alpha} \\ \text{s.t.} & \mu_p \geq \tilde{\mu}_p, \quad \boldsymbol{\alpha}'\mathbf{e} = 1 \end{cases} \quad \text{or} \quad \begin{cases} \max\limits_{\{\alpha\}} & \mu_p = \boldsymbol{\alpha}'\boldsymbol{\mu} \\ \text{s.t.} & \sigma_p \leq \tilde{\sigma}_p, \quad \boldsymbol{\alpha}'\mathbf{e} = 1. \end{cases}$$

Proposition: When there are no restrictions on the portfolio weights, the weights of the **minimum variance portfolio (MVP)** for a required return $\tilde{\mu}_p$ are:

$$\boldsymbol{\alpha}^* = \boldsymbol{\Lambda}_1 + \boldsymbol{\Lambda}_2\,\tilde{\mu}_p,$$

where $\boldsymbol{\Lambda}_1 = \frac{\boldsymbol{\Sigma}^{-1}}{D}(B\mathbf{e} - A\boldsymbol{\mu}), \qquad \boldsymbol{\Lambda}_2 = \frac{\boldsymbol{\Sigma}^{-1}}{D}(C\boldsymbol{\mu} - A\mathbf{e})$

with $A = \mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}, \quad B = \boldsymbol{\mu}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}, \quad C = \mathbf{e}'\boldsymbol{\Sigma}^{-1}\mathbf{e}, \quad D = BC - A^2.$

The collection of MVPs for various $\tilde{\mu}_p$ gives the **mean-variance efficient frontier.**

The efficient frontier is given by the relation:

$$\tilde{\mu}_p = \frac{A}{C} + \sqrt{\frac{D}{C}\left(\tilde{\sigma}_p^2 - \frac{1}{C}\right)}.$$

Remarks:

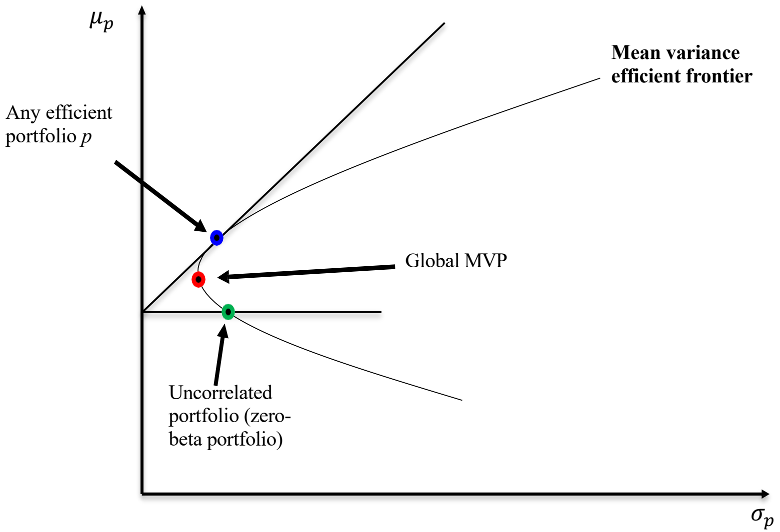
- Any portfolio of MVPs is also an MVP.
- The covariance between two efficient MVPs is always positive.
- The **Global Minimum Variance Portfolio** (Global MVP) is given by

$$\boldsymbol{\alpha}_g = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{e}}{C}, \qquad \mu_g = \frac{A}{C}, \qquad \sigma_g^2 = \frac{1}{C}.$$

- The covariance of any asset or portfolio return R_p with the Global MVP is

$$\mathrm{Cov}(R_g, R_p) = \frac{1}{C}.$$

Mean-Variance Case (Markowitz, 1952) – Mean-variance efficient frontier



Mean-Variance Case (Tobin, 1958) – Solution with a risk-free asset

There is a risk-free asset, which the investor can borrow or lend with unlimited amount. The solution is the portfolio weight $\boldsymbol{\alpha}^*$ that maximizes:

$$\mu_p - \frac{\lambda}{2}\sigma_p^2 = \left[\boldsymbol{\alpha}'\boldsymbol{\mu} + (1 - \boldsymbol{\alpha}'\mathbf{e})R_f\right] - \frac{\lambda}{2}\boldsymbol{\alpha}'\boldsymbol{\Sigma}\boldsymbol{\alpha}.$$

Proposition: If there is a risk-free asset, the weights of the **optimal portfolio** are:

$$\boldsymbol{\alpha}^* = \frac{1}{\lambda}\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - R_f\mathbf{e}) \qquad \text{in risky assets}$$

$$(1 - \mathbf{e}'\boldsymbol{\alpha}^*) \qquad \text{in the risk-free asset.}$$

The sum of the $\boldsymbol{\alpha}^*$ does not necessarily sum to 1, because the investor can hold some amount of risk-free asset. The set of all portfolios when λ varies is called the **Capital Allocation Line**.

Remark: If there is one risky asset only (μ_m, σ_m^2), then $\boldsymbol{\alpha}^* = (\mu_m - R_f)/(\lambda\sigma_m^2)$.

Mean-Variance Case (Tobin, 1958) – Alternative derivation

An investor minimizing the portfolio variance subject to a return constraint will find the same solution:

$$\begin{cases} \min\limits_{\{\boldsymbol{\alpha}\}} & \sigma_p^2 = \boldsymbol{\alpha}'\boldsymbol{\Sigma}\boldsymbol{\alpha} \\ \text{s.t.} & \mu_p = \boldsymbol{\alpha}'\boldsymbol{\mu} + (1 - \mathbf{e}'\boldsymbol{\alpha})R_f \geq \tilde{\mu}_p \end{cases} \quad \text{or} \quad \begin{cases} \max\limits_{\{\boldsymbol{\alpha}\}} & \mu_p = \boldsymbol{\alpha}'\boldsymbol{\mu} + (1 - \mathbf{e}'\boldsymbol{\alpha})R_f \\ \text{s.t.} & \sigma_p = \sqrt{\boldsymbol{\alpha}'\boldsymbol{\Sigma}\boldsymbol{\alpha}} \leq \tilde{\sigma}_p. \end{cases}$$

Proposition: If there is a risk-free asset, the weights of the risky assets in the **optimal portfolio** for a required return $\tilde{\mu}_p$ are:

$$\boldsymbol{\alpha}^* = \frac{\tilde{\mu}_p - R_f}{(\boldsymbol{\mu}^e)'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e = \frac{\gamma}{2}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e,$$

where $\boldsymbol{\mu}^e = \boldsymbol{\mu} - R_f\mathbf{e}$ is the vector of excess returns and γ is the Lagrange multiplier.

Tobin's Two-fund Separation Theorem: Every mean-variance efficient portfolio is a combination of the risk-free asset and the **tangency portfolio** with weight:

$$\boldsymbol{\alpha}_T = \frac{\boldsymbol{\alpha}^*}{\mathbf{e}'\boldsymbol{\alpha}^*} = \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e}{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e}, \qquad \text{where} \quad \boldsymbol{\mu}^e = \boldsymbol{\mu} - R_f\mathbf{e}.$$

The tangency portfolio is also characterized by:

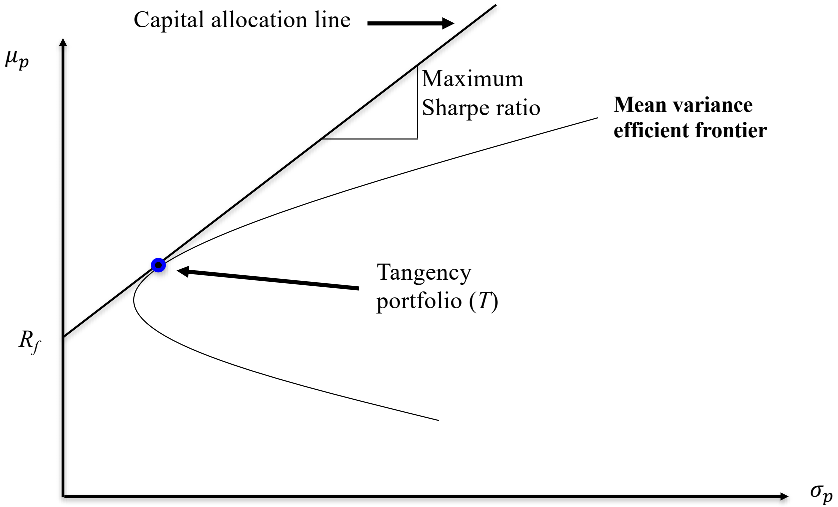
$$\mu_T - R_f = \boldsymbol{\alpha}_T'\boldsymbol{\mu}^e = \frac{(\boldsymbol{\mu}^e)'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e}{\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e} \qquad \text{and} \qquad \sigma_T^2 = \boldsymbol{\alpha}_T'\boldsymbol{\Sigma}\boldsymbol{\alpha}_T = \frac{(\boldsymbol{\mu}^e)'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e}{(\mathbf{e}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}^e)^2}.$$

The risk-adjusted performance (Sharpe ratio) is

$$\frac{\mu_T - R_f}{\sigma_T} = \sqrt{(\boldsymbol{\mu} - R_f\mathbf{e})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - R_f\mathbf{e})}.$$

Remark: The tangency portfolio is the risky portfolio with the maximum Sharpe ratio.

Mean-Variance Case (Tobin, 1958) – Solution with a risk-free asset



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Market Equilibrium and CAPM

Previously, expected excess returns were taken as given and known.

In fact, expected excess returns $\boldsymbol{\mu}^e$ should result from the confrontation of supply and demand, i.e., it should be consistent with an equilibrium model.

The **CAPM** provides an equilibrium model explaining where expected excess returns come from.

It has been extended to allow for multiple factors.

Market Equilibrium and CAPM: Assumptions

1. Investors are price-takers (believe that security prices are unaffected by their own trades)
2. All investors plan for one identical holding period (myopic).
3. Universe: publicly traded assets + risk-free borrowing/lending at R_f.
4. No taxes and no transaction costs.
5. All investors are rational mean-variance optimizers (they all use the Markowitz/Tobin portfolio selection model above)
6. Homogeneous expectations: same expected returns and covariance matrix; different risk aversion allowed.

Market Equilibrium and CAPM: Results

- **R1.** All investors hold the same market portfolio M (market-value-weighted portfolio of all securities).

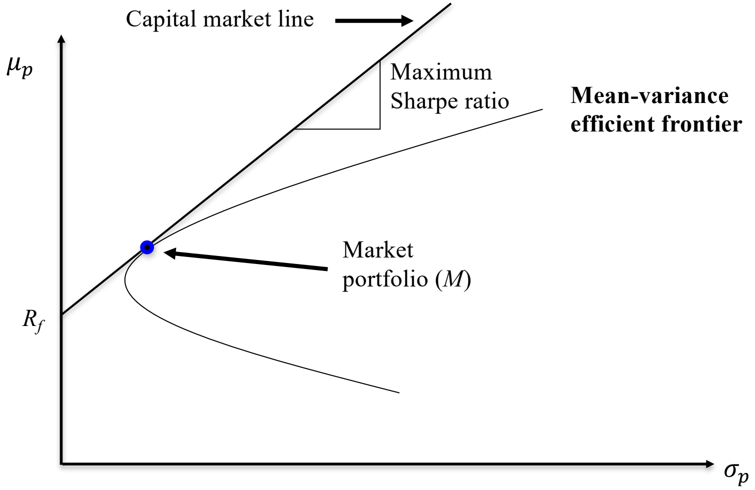
- **R2.** M lies on the efficient frontier and is the tangency portfolio. The line from R_f through M is the best attainable capital allocation line (CML).

- **R3.** The market risk premium is proportional to market risk and representative investor risk aversion.

$$\mathbb{E}_t[R_{m,t+1}] - R_{f,t} = \lambda \sigma_m^2 \qquad (\alpha^* = 1)$$

- **R4.** The risk premium on individual assets is proportional to the risk premium on the market portfolio, M, and the beta coefficient of the security relative to the market.
  - There is only one risk factor for the assets: their correlation with the market portfolio (with return $R_{m,t+1}$).
  - This correlation is measured by the beta parameter: $\beta_i = \frac{\mathrm{Cov}(R_{i,t+1}, R_{m,t+1})}{\mathrm{Var}(R_{m,t+1})} = \frac{\mathbf{e}_i' \boldsymbol{\Sigma} \boldsymbol{\alpha}}{\boldsymbol{\alpha}' \boldsymbol{\Sigma} \boldsymbol{\alpha}}$ where $\mathbf{e}_i$ is the i-th canonical vector (1 at position i, 0 otherwise)

Market Equilibrium and CAPM



Risk premium and beta (CAPM intuition)

At equilibrium, the expected excess return (or risk premium) of individual asset i is "price times quantity of risk":

$$\underbrace{\mathbb{E}_t[R_{i,t+1}] - R_{f,t}}_{\substack{\text{Expected excess} \\ \text{return of asset } i}} = \underbrace{\frac{\mathbb{E}_t[R_{m,t+1} - R_{f,t}]}{\sigma_m}}_{\text{Price of risk}} \times \underbrace{\frac{\mathrm{Cov}(R_{i,t+1}, R_{m,t+1})}{\sigma_m}}_{\substack{\text{Systematic risk} \\ \text{of asset } i}}$$

or equivalently

$$\mathbb{E}_t[R_{i,t+1}] - R_{f,t} = \frac{\mathrm{Cov}(R_{i,t+1}, R_{m,t+1})}{\mathrm{Var}(R_{m,t+1})} \mathbb{E}_t[R_{m,t+1} - R_{f,t}] = \beta_i \, \mathbb{E}_t[R_{m,t+1} - R_{f,t}] \, .$$

CAPM: The risk premium of asset i is equal to its beta $\times$ the expected excess return of the market portfolio. Assets with high systematic risk (high β) offer high expected return.

CAPM and Multi-Factor Models

- CAPM: Sharpe (1964), Lintner (1965)
- **Limitations to CAPM:**
  - Market portfolio is not directly observable
  - Empirical evidence suggests additional return drivers
- **Fama-French three-factor model**
  - Market beta
  - Size
  - Book-to-market

Fund evaluation often uses the adjusted CAPM regression:

$$R_{p,t+1} - R_{f,t} = \alpha_p + \beta_p \left(R_{b,t+1} - R_{f,t} \right) + \varepsilon_{p,t+1}.$$

where $R_{p,t+1}$ is the fund return and $R_{b,t+1}$ is the return of the benchmark of the fund

Passive management ≈ beta exposure versus **active management ≈ alpha**

Objectives of the lecture

- What Is Sustainable Finance?
- Portfolio Optimization
- Capital Asset Pricing Model
- **Market Efficiency**

Efficient Market Hypothesis (EMH)

In an **efficient** market, stock prices fully reflect available information (Fama, 1965).

Market efficiency does not mean returns are zero; it means there is no profit beyond the **normally required return**.

Normal returns can be defined by an asset pricing model (e.g., CAPM).

$$\mathbb{E}_t[R_{i,t+1}] = R_{f,t} + \beta_i \mathbb{E}_t(R_{m,t+1} - R_{f,t}))$$

Abnormal returns are:

$$\varepsilon_{i,t+1} = R_{i,t+1} - \mathbb{E}_t[R_{i,t+1}]$$

Implications:

- Public information is incorporated quickly, so knowing it at release does not generate abnormal profit
- Firms should receive fair value when issuing securities; fooling investors should not systematically work

Implications of Market Efficiency: Time-Series

- **Stock prices** (including dividends) follow a **random walk** with positive trend (risk compensation).

- **Expected excess returns** (risk premia) are determined by equilibrium models (e.g., CAPM).

$$\underbrace{R_{i,t+1} - R_{f,t}}_{\text{Excess return}} = \underbrace{\mathbb{E}_t[R_{i,t+1} - R_{f,t}]}_{\text{Expected excess return}} + \underbrace{\varepsilon_{i,t+1}}_{\text{Abnormal return}}$$

- As information flow is random, **abnormal returns are unpredictable** conditional on a given information set.

- **No free lunch**: systematic abnormal returns should not be attainable.

- Timing the market does not work.

Implications of Market Efficiency: Investment

- **Fundamental analysis** should not systematically work if information is public and efficiently priced.
- Under EMH, **active management** is often wasted effort and costly.
- **Passive management** is the benchmark strategy:
  - buy-and-hold,
  - highly diversified portfolio (market portfolio),
  - index funds tailored to investor constraints.

Efficient Market Hypothesis (EMH): Evidence

- Weak-form EMH is not rejected by the data.
- Professional manager performance is broadly consistent with semi-strong efficiency: deviations from the market are within statistical uncertainty margins.
- Markets are very efficient, but especially diligent/intelligent/creative investors may outperform on average.
- Private information is valuable, but generally prohibited to trade on (insider trading). What about high-frequency trading?

From Portfolio Management to Benchmarking and Active Management

- In this lecture, we focused on **portfolio management**: mean-variance optimization, CAPM, and market efficiency.
- In practice, most mandates are **benchmark-oriented**: a **passive** benchmark plus **active** deviations.
- In Lecture 2, we move from **passive** to **active** management and detail issues in estimating expected returns.

End of Lecture 1